TASFIA JANNAT TASFI

Natural Language Processing | LLM Evaluation | Human–AI Interaction

Rajshahi, Bangladesh | tj2003115@gmail.com
linkedin.com/in/tasfia-jannat-tasfi | shion115.github.io/website

Research Profile

I am a Computer Science and Engineering graduate with research experience in **NLP, large language model (LLM) evaluation, and AI-enabled software systems**. My work includes evaluating reasoning failures in multilingual LLMs and building an NLP-backed application that communicates model uncertainty and supports human decision-making. I am interested in extending these skills to **AI-assisted software engineering**, particularly the empirical evaluation of LLM-based developer tools, the analysis of natural-language software artifacts, and reliable human–AI workflows.

Research Interests: AI for Software Engineering; NLP and LLMs; Empirical Evaluation of AI Systems; Software Reliability; Human–AI Interaction; Intelligent Developer Tools

Education

B.Sc. in Computer Science and Engineering **11 Aug. 2026**
Rajshahi University of Engineering & Technology (RUET) *Rajshahi, Bangladesh*

CGPA: 3.77/4.00 | **Program:** BAETE-accredited CSE program

Research Experience

Undergraduate Thesis: Designing and Evaluating Explanation Interfaces for AI-Assisted Medical Misinformation Detection

Natural Language Processing | Reliable AI Predictions | Experimental Evaluation | Human–AI Interaction

- Developed and evaluated an **RoBERTa-based NLP system** using a multi-source corpus of **17,611 samples**, incorporating probability calibration and conformal selective prediction to better distinguish automated decisions from cases requiring human review.
- Achieved **0.9806 macro-F1** on automatically handled cases at **91.56% automation coverage**; evaluated model predictions and uncertainty as part of an end-to-end research system.
- Designed and implemented **four web-based interfaces** with **Next.js and FastAPI** to communicate predictions, linguistic cues, uncertainty, evidence, and sources; compared how alternative representations affected the interpretation of model outputs.
- Conducted a controlled study with **60 participants and 1,200 judgment trials**. The most informative interface increased judgment accuracy from **73.7% to 90.0%** and reduced reliance on incorrect AI advice from **77.8% to 37.8%**.

Publications

Nahid Montasir Rifat, Tasfia Jannat Tasfi, Foyez Ahmed Dewan, and Barshon Sen. *Evaluating Creative Reasoning and Lateral Thinking in Instruction-Tuned LLMs for Bengali.* 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), 2026.
DOI: https://doi.org/10.1109/QPAIN69676.2026.11546604

Developed a parallel Bangla–English reasoning benchmark for instruction-tuned LLMs and examined model failures involving mistranslation, cultural mismatch, and logical inconsistency.

Jabunnesa Jahan Sara, Shahriar Rizvi, and Tasfia Jannat Tasfi. *Beyond Artifacts: Semantic Deepfake Detection and Digital Provenance Tracking with Multimodal Large Language Models.* 2025 28th International Conference on Computer and Information Technology (ICCIT), 2025.
DOI: https://doi.org/10.1109/ICCIT68739.2025.11490141

Developed a multimodal LLM-based approach to semantic deepfake analysis and web-supported provenance tracing, focusing on verification beyond visual artifact detection.

Selected Achievements

- **1st Place, UIHP 5th Cohort (WayMakers)** — recognized for startup innovation, market validation, and MVP development.
- **Champion**, Huawei Seeds for the Future Tech-Business Competition, 2024.
- **Champion**, Hult Prize at RUET — Startup Innovation, 2023.
- **Government General Scholarship** — awarded for all four years of undergraduate study.
- **Finalist**, International Youth Math Challenge — Top 20% globally, 2024.
- **70% Merit Scholarship**, ISCEA PTAK Prize, 2024.
- **Government Talent Pool Scholarship** — awarded for academic excellence in high school.

Selected AI & Software Projects

BongoSeek — LLM-Powered Conversational Search. Developed an LLM-based conversational search application with context-aware query handling and dialogue memory for coherent multi-turn responses; implemented and integrated application components for interactive use.
Code: github.com/Shion115/Bongo_Seek

EarlyMind — AI-Assisted Child Development Interface. Built an interactive web application for child-development assessment activities, with adaptive dashboards and progress feedback informed by cognitive-load considerations.
Code: github.com/Shion115/EarlyMind

Professional & Leadership Experience

Co-Founder, Delibro **Nov. 2024 – Present**
Community-Based Delivery Software Platform
- Co-developed a community-oriented delivery platform, contributing to application design, prototyping, product development, and user validation. Received support through the **World Bank-supported University Innovation Hub**.

Official Ambassador **Jun. 2025 – Present**
International Computer Science Competition
- Led programming outreach and mentorship. **Profile:** icscompetition.org/en/ttasfi

Industrial Trainee **Mar. 2025**
BRAC IT
- Worked on front-end interface design and web-development workflows.

App Developer **Jun. 2022 – Jan. 2023**
bdapps
- Developed and deployed PHP-based applications involving interface implementation, backend integration, and deployment.

Research & Technical Skills

NLP & LLMs	Transformers, RoBERTa, Hugging Face Transformers, LLM Evaluation, Multilingual Benchmarking
Evaluation	Controlled Experiments, Model Evaluation, Error Analysis, Uncertainty Calibration, Conformal Prediction
Development	Python, C++, JavaScript, PHP, Next.js, FastAPI, Flask, HTML/CSS, Git
Research Tools	scikit-learn, pandas, Jupyter Notebook, Google Colab, Figma, UX Prototyping

References

Barshon Sen
Assistant Professor, Department of CSE, RUET
Undergraduate Thesis Supervisor
barshon.sen@cse.ruet.ac.bd

Md. Azmain Yakin Srizon
Assistant Professor, Department of CSE, RUET
Academic Advisor
azmainsrizon@cse.ruet.ac.bd